

Contributing to SOLVE-IT - A Guide for Researchers

If you are a digital forensics researcher then there is a good chance that you are carrying out and publishing work that could be added to the [SOLVE-IT knowledge base](#). This short document should provide you with the details needed to add your work to the knowledge base. Page 2 contains a [Quick Start Guide](#), with [more detail](#) provided in the later sections.

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Why add your work?

Visibility

The DFPulse 2024 Practitioner Survey (Hargreaves et al. 2024) revealed that the links between academic work and digital forensics practitioners are very poor. One of the benefits of SOLVE-IT is that it is rooted in tangible techniques that are usable in investigations. Linking your work in the specific context of a digital forensic technique makes your work easier to find. Distilling it down to the fields captured by the entries for techniques, weaknesses or mitigations makes the application of your work obvious, while providing a means to link to the original publication.

Extending thinking on the topic

Some researchers have reported that the act of distilling their work into a SOLVE-IT technique and enumerating weaknesses and mitigations has improved their understanding of their topic or process:

- *Considering timeline-based techniques in terms of SOLVE-IT organization shows some distinct processes such as timeline generation (T1052), timeline analysis (T1086), and time-based event reconstruction (T1117).*
- *Analysing T1087: Location-based event reconstruction revealed that “Visualisation of geolocation information” is a distinct technique (T1115), with its own weaknesses and mitigations to explore further. This was added to the knowledge base.*

SOLVE-IT Contribution Quick Start Guide

1. Decide what you are adding

Contribution Type*	When to Use	Example
Technique	Your work describes a <i>method</i> or process to achieve a forensic objective.	File Carving
Weakness	Your work identifies a potential <i>problem</i> with using a technique.	Incorrect attribution of salvaged content
Mitigation	Your work offers a <i>solution</i> to a known weakness.	Use digital stratigraphy to attribute data

* Note that a single research paper may result in multiple contributions

2. The contribution workflow



3. To develop your content for submission use the TRWM checklist

(Technique → Results → Weaknesses → Mitigations)

- **Techniques:** name, description, details, examples.
- **Results:** What outputs are there from the technique? (map to CASE classes if possible).
- **Weaknesses:** Use ASTM E3016-18 error categories to systematically enumerate weaknesses linked to each result: INCOMP, INAC-EX, INAC-AS, INAC-ALT, INAC-COR, MISINT (guidance is provided in project documentation).
- **Mitigations:** Think about what can be done to avoid, or minimise the impact of the weaknesses.

Submit your techniques, weaknesses, or mitigations via the project issue tracker using one of the provided templates.

4. JSON Template for a Technique

Encode your content in JSON form and submit a pull request.

```
{
  "id": "TXXXX",
  "name": "Technique Name",
  "description": "Short, precise definition.",
  "synonyms": [],
  "details": "Longer explanation with context.",
  "subtechniques": [],
  "examples": ["Tool1", "Tool2"],
  "weaknesses": ["WXXXX", "WYYYY"],
  "CASE_output_classes": ["observable:ClassName"],
  "references": ["Author, Year, Title, Journal, DOI"]
}
```

5. Quick Links

- **SOLVE-IT Repo:**
<https://github.com/SOLVE-IT-DF/solve-it>
- **SOLVE-IT Issue tracker:**
<https://github.com/SOLVE-IT-DF/solve-it/issues>
- **SOLVE-IT Contributing.md:**
<https://github.com/SOLVE-IT-DF/solve-it/blob/main/CONTRIBUTING.md>
- **Education Repo (TRWM examples):**
<https://github.com/SOLVE-IT-DF/solve-it-education>
- **CASE Ontology Classes:**
<https://ontology.caseontology.org/documentation/entities-az.html>

Extended Guidance

Overview

Within SOLVE-IT there are four main concepts: *objectives*, *techniques*, *weaknesses* and *mitigations*. These are:

- **Objectives** - based on ATT&CK tactics, objectives are "the goal that one might wish to achieve in a digital forensic investigation", e.g. acquire data, or extract information from a file system.
- **Techniques** - "how one might achieve an objective in digital forensics by performing an action", e.g. for the objective of 'acquire data', the technique 'create disk image' could be used.
- **Weaknesses** - these represent potential problems resulting from using a technique. They are classified according to the error categories in ASTM E3016-18, the Standard Guide for Establishing Confidence in Digital and Multimedia Evidence Forensic Results by Error Mitigation Analysis.
- **Mitigations** - something that can be done to prevent a weakness from occurring, or to minimise its impact.

Typically, much applied digital forensics research develops both mitigations and techniques in SOLVE-IT terminology, but research into that technique may have started with awareness of a problem in digital forensics that needed to be solved. Again, in SOLVE-IT terminology, perhaps this was a weakness in an existing technique, leading to the idea for a mitigation, which developed enough to be indexed as a technique in its own right.

Examples

Example 1

For example, the work by Vanini et al. (2024) which describes the potential problem of using the clock offset recovered from the BIOS/EFI of a computer to apply a time skew universally over all recovered timestamps. The issue is that the system clock could have had various skews at different points in time, either naturally, or as a result of clock tampering. The work on Time Anchors in Vanini et al (2024) explores the technique where artefacts can be extracted and examined that simultaneously contain both a timestamp based on the local system clock, and a timestamp from an external source. These offer the potential to determine different timeskews at multiple points in the device's history.

This research therefore identifies a weakness in an existing technique, determines a potential mitigation, and develops it into a technique. This leads to:

- W1148: Incorrect time skew applied to a timeline entry (in T1052)
- W1149: Failure to determine clock accuracy at time of inferred event (in T1117)
- M1103: Use 'Time Anchors' to assess clock accuracy at time of inferred event

These can be found in the SOLVE-IT knowledge base. This mitigation in turn could be developed into a distinct technique within SOLVE-IT.

Example 2

As another example, Schneider et al. (2024) considers an issue identified in Schneider et al. (2021), where for removable media, due to reuse of recycled storage chips, it is not possible to safely say that data recovered from non-allocated space originated from the current file system. The more recent paper offers an application of digital stratigraphy to determine if a carved file from non-allocated space is likely to belong to the current file system or a previous one. This can be developed into the weakness, mitigation and technique of:

- W1106: Incorrect attribution of salvaged content to a current file system rather than a previous one (in T1064)
- M1061: Use digital stratigraphy to attempt to attribute data within a specific file system

Example 3

Hargreaves et al (2025) looks at the forensic artefacts left by the Replika AI companion app, but through a TRWM analysis (see [later](#)) based on the results, was able to produce the technique **T1128: AI companion app examination**, which also added 27 new weaknesses, 18 new specific mitigations, and linked to 3 existing ones.

Planning your contribution

The overall process for adding content into SOLVE-IT is to raise an issue in the GitHub project [issue tracker](#) using one of the templates, request review, and then, after a response, implement the changes in the JSON backend knowledge base content.

To raise the issue you will need to understand what you are going to propose changing, which will be some combination of:

- Add technique
- Update technique
- Add weakness
- Update weakness
- Add mitigation
- Update mitigation

To get started, it is recommended to review the current SOLVE-IT knowledge base content, either in text form in the `.repo_info` folder (techniques.txt, weaknesses.txt, mitigations.txt), or the raw JSON, or by downloading the latest Excel spreadsheet representation.

It may help to start by determining which of the objectives your technique is likely to relate to, e.g. 'Acquire data', 'Extract artifacts stored by applications', or 'Locate potentially relevant content', etc. This will help you to look at related techniques as a guide. This would also apply if you have a weakness to add - think about the sort of technique that would have such a weakness and see if any already exist. The Excel workbook version also has a worksheet 'Weaknesses' which maps each weakness to any technique that refers to it.

The **SOLVE-IT MCP Server Project** (https://github.com/CKE-Proto/solve_it_mcp) may help with interacting with the knowledge base using natural language. In the background the MCP server can retrieve all entities by ID but also search for keywords derived from your query and return natural language results with direct references into the data in the SOLVE-IT knowledge base. Alternatively, the plaintext techniques.txt, weaknesses.txt and mitigations.txt can be uploaded to other services such as ChatGPT for a quick LLM based review of the content, but is not as effective as the MCP server approach.

To assist further, some questions to consider when placing your work in SOLVE-IT:

Does this work *identify* a problem in digital forensics?

- If yes then you have potentially identified a **weakness** that can be added to the knowledge base.
 - a) Is that problem an existing **weakness** of existing technique? If yes:
 - You can check the description and details and submit an issue for **Update Weakness** if any changes are needed in light of your work, including a citation for your research.
 - b) Is that problem not listed as a weakness for the existing relevant technique? If yes:
 - You can submit an issue to **Propose New Weakness** and provide details.

c) Is the problem associated with a technique that is not yet indexed in SOLVE-IT? If yes:

- You will first need to add the technique so that the weakness can be attached to it. You can submit an issue for **Propose New Technique**, and details of the weakness(es) can be provided alongside.

Does this work solve a problem in digital forensics?

By solving it you have also by default identified it (see above). Furthermore you can consider:

a) Can your solution be captured in one line and a link to your research? If yes:

- You can submit an issue for **Propose New Mitigation** which needs only a very brief description and citation.

b) Does your solution need extended descriptions, details and should the weaknesses of the approach also be described. If yes:

- You can first **Propose New Technique**, and provide all the details needed. This can include potential weaknesses and mitigations.
- Then if the technique also mitigates existing weaknesses, you will need to also **Propose New Mitigation**, which describes the use of your technique in a specific context (i.e. of mitigating a weakness). That mitigation can be described briefly but can link to your research in the citation, and the field 'technique' can link to the ID of the technique you just created.

Types of Issue Submissions

For many types of issue tracker submissions there are variations listed. These are:

- LITE - minimal details to capture the idea of the technique or weakness
- STANDARD - all details needed in a plain text format for techniques, weaknesses and mitigations.
- TRWM - applies to techniques only, and involves a full systematic analysis of the technique and enumerated weaknesses and mitigations

Depending on which issue type you submit, you may be asked for more details but getting started is the most important thing, so feel free to begin with the LITE submission; it was created to make submissions easy to get started!

A Full TRWM (Technique Results Weakness Mitigations) Analysis

The TRWM submission is the opposite end of the scale to the LITE submissions. As mentioned above, submitting using any of the issue types is fine, but as a digital forensic researcher, given your in depth knowledge of the technique you are adding, the most appropriate is the full TRWM

analysis which asks for the most details and provides a structured method for enumerating weaknesses and mitigations.

An example exercise that walks through the analysis process is available in the SOLVE-IT education repository (<https://github.com/SOLVE-IT-DF/solve-it-education> - *Weakness Enumeration with the TRWM process*). You can use that exercise to help perform the analysis of your technique. A summary of the approach is described below:

- **Technique:** Provide the details of the technique, including name, and description
- **Results:** Consider what the type of outputs of the technique are. These can be mapped against current or potential future CASE classes from the [ontology](#).
- **Weaknesses:** For each type of technique result, use the ASTM error classifications¹ to consider what each type of error may look like for that particular result. These are weaknesses.
- **Mitigations:** For each weakness you can then consider what mitigations are possible.

The TWRM helper workbooks are designed to walk you through this formal enumeration process. The latest version is linked from the '[TRWM Explained](#)' section of the education repository ready for duplication and completion.

Implementing the Contribution

The repository has a document ([CONTRIBUTING.md](#)) which provides detail on this, but a summary is provided below.

Submitting an Issue

- Visit the SOLVE-IT issue tracker
- Select one of the templates (LITE, STANDARD, TRWM) and complete the details for review.
- After submission someone will review your proposal, perhaps suggest improvements, then assign you SOLVE-IT technique, weakness and mitigation IDs to use for your implementation.

JSON updates and Pull Requests

- Follow the [GitHub Flow](#) workflow
 - Fork the repository so you have your own copy.
 - Create a new [branch](#) from `main`. To streamline the creation of merge requests, start your Git branch name with the issue number, followed by a hyphen.
 - Make your changes and commit them with descriptive commit messages.
 - Submit a pull request to the main SOLVE-IT repository's `main` branch.

¹ Incompleteness, Inaccuracy (existence), Accuracy (association), Inaccuracy (alteration), Inaccuracy (corruption), Misinterpretation. Referred to as INCOMP, INAC-EX, INAC-AS, INAC-ALT, INAC-COR, and MISINT.

Other tips

- Check out the repository using a tool such as [GitHub Desktop](#).
- When making your edits to the repository, the JSON files that need to be edited are in the /data folder.
- To update a technique, locate the corresponding json file in the /data/techniques folder. Update that JSON file with the relevant information.
- New techniques need to be added (by ID) to solve-it.json under the correct objective.
- New techniques can be added by creating a new json file in the same structure. Technique T1000.json provides a template.
- You can reference weaknesses and mitigations, either existing ones in the /data/weaknesses or /data/mitigations folders, or create new ones.
- Make sure you re-run `generate_excel_fromkb.py` to ensure your data is integrated and linked correctly before submitting a pull request.

More information is available in [CONTRIBUTING.md](#)

Workflow based research output

Some published research does not map to a single technique. In this case, consider mapping your work as a workflow, made up of multiple techniques (as per the [disk imaging example](#) in the examples repository). It may be that some parts of your work are covered by existing techniques, but details of the workflow need to be added to SOLVE-IT to fully capture the workflow.

Workflows are not captured in the SOLVE-IT knowledge base but consider submitting to the [SOLVE-IT examples repository](#).

Further Information

- See [CONTRIBUTING.md](#) for more details
- The [SOLVE-IT education repository](#) has a specific exercise on Weakness enumeration using TRWM.

References

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